Bibliography

Abdolmaleki, S., & Abadeh, M. S. (2020). Brain mr image classification for adhd diagnosis using deep neural networks. *2020 International Conference on Machine Vision and Image Processing (MVIP)*, 1–5. https://doi.org/10.1109/MVIP49855.2020.9116877

Achalia, R., Sinha, A., Jacob, A., Achalia, G., Kaginalkar, V., Venkatasubramanian, G., & Rao, N. P. (2020). A proof of concept machine learning analysis using multimodal neuroimaging and neurocognitive measures as predictive biomarker in bipolar disorder. *Asian Journal of Psychiatry*, *50*, 101984.

Asraf, A., & Islam, Z. (2021). *Covid19, pneumonia and normal chest x-ray pa dataset* (Version 1). Mendeley Data. https://doi.org/10.17632/mxc6vb7svm.1

Bae, J.-H., Yu, G.-H., Lee, J.-H., Vu, D. T., Anh, L. H., Kim, H.-G., & Kim, J.-Y. (2022). Superpixel image classification with graph convolutional neural networks based on learnable positional embedding. *Applied Sciences*, *12*(18), 9176. https://doi.org/10.3390/app12189176

Barstugan, M., Ozkaya, U., & Ozturk, S. (2021). Coronavirus (covid-19) classification using ct images by machine learning methods. *Proceedings of RTA-CSIT 2021*, *2872*, 29–35. https://ceur-ws.org/Vol-2872/paper05.pdf

Behrendt, F., et al. (2023). A systematic approach to deep learning-based nodule detection in chest radiographs. *Scientific Reports*, *13*, 10353. https://doi.org/10.1038/s41598-023-37270-2

Besenczi, R., Tóth, J., & Hajdu, A. (2016). A review on automatic analysis techniques for color fundus photographs. *Computational and Structural Biotechnology Journal*, *14*, 371–384. https://doi.org/10.1016/J.CSBJ.2016.10.001

Bhagwat, N., Viviano, J. D., Voineskos, A. N., & Chakravarty, M. M. (2018). Modeling and prediction of clinical symptom trajectories in alzheimer's disease using longitudinal data. *PLOS Computational Biology*, *14*, e1006376. https://doi.org/10.1371/JOURNAL.PCBI.1006376

Biloborodova, T., Brosnan, B., Skarga-Bandurova, I., & Strauss, D. J. (2023). Generalization ability in medical image analysis with small-scale imbalanced datasets: Insights from neural network learning. In N. Moniz, Z. Vale, J. Cascalho, C. Silva, & R. Sebastião (Eds.), *Progress in artificial intelligence* (pp. 234–246). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-49011-8_19

Brugnara, G., Neuberger, U., Mahmutoglu, M. A., Foltyn, M., Herweh, C., Nagel, S., Schönenberger, S., Heiland, S., Ulfert, C., Ringleb, P. A., et al. (2020). Multimodal predictive modeling of endovascular treatment outcome for acute ischemic stroke using machine-learning. *Stroke*, *51*(12), 3541–3551.

Brussee, S., Buzzanca, G., Schrader, A. M., & Kers, J. (2025). Graph neural networks in histopathology: Emerging trends and future directions. *Medical Image Analysis*, *101*, 103444. https://doi.org/https://doi.org/10.1016/j.media.2024.103444

Burwinkel, H., Kazi, A., Vivar, G., Albarqouni, S., Zahnd, G., Navab, N., & Ahmadi, S. A. (2019). Adaptive image-feature learning for disease classification using inductive graph networks. *International Conference on Medical Image Computing and Computer-Assisted Intervention*, 640–648. https://doi.org/10.1007/978-3-030-32226-7_71

Cai, H., Gao, Y., & Liu, M. (2023). Graph transformer geometric learning of brain networks using multimodal mr images for brain age estimation. *IEEE Transactions on Medical Imaging*, *42*(2), 456–467. https://doi.org/10.1109/TMI.2022.3222093

Chen, C.-F. R., Fan, Q., & Panda, R. (2021). Crossvit: Cross-attention multi-scale vision transformer for image classification. *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, 347–356. https://doi.org/10.1109/ICCV48922.2021.00041

Chen, Q., Huang, G., & Wang, Y. (2022). The Weighted Cross-Modal Attention Mechanism with Sentiment Prediction Auxiliary Task for Multimodal Sentiment

Analysis. *IEEE/ACM Transactions on Audio Speech and Language Processing*, *30*, 2689–2695. https://doi.org/10.1109/TASLP.2022.3192728

Chen, T., Hong, R., Guo, Y., Hao, S., & Hu, B. (2023). Ms2-gnn: Exploring gnn-based multimodal fusion network for depression detection. *IEEE Transactions on Cybernetics*, *53*, 7749–7759. https://doi.org/10.1109/TCYB.2022.3197127

Chen, X., Wang, S., & Kong, W. (2025). Multimodal data fusion for alzheimer's disease based on dynamic heterogeneous graph convolutional neural network and generative adversarial network. *Array*, *26*, 100415. https://doi.org/10.1016/j.array.2025.100415

Cosma, R. A., Knobel, L., van der Linden, P., Knigge, D. M., & Bekkers, E. J. (2023). Geometric superpixel representations for efficient image classification with graph neural networks. *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 109–118. https://doi.org/10.1109/ICCVW60793.2023.00018

Coupeau, P., Fasquel, J.-B., & Dinomais, M. (2024). On the use of GNN-based structural information to improve CNN-based semantic image segmentation. *Journal of Visual Communication and Image Representation*, *101*, 104167. https://doi.org/10.1016/j.jvcir.2024.104167

Cui, C., Yang, H., Wang, Y., Zhao, S., Asad, Z., Coburn, L. A., Wilson, K. T., Landman, B. A., & Huo, Y. (2022). Deep multimodal fusion of image and non-image data in disease diagnosis and prognosis: A review. *arXiv preprint*, *arXiv:2203.15588*. https://arxiv.org/abs/2203.15588

Decuyper, M., Bonte, S., Deblaere, K., & Holen, R. V. (2021). Automated mri based pipeline for segmentation and prediction of grade, idh mutation and 1p19q co-deletion in glioma. *Computerized Medical Imaging and Graphics*, *88*, 101831. https://doi.org/10.1016/J.COMPMEDIMAG.2020.101831

Deng, Q., Wu, L., Su, K., Wu, W., Li, Z., & Duan, W. (2024). Hierarchical Fusion Framework for Multimodal Dialogue Response Generation, 1–8. https://doi.org/10.1109/IJCNN60899.2024.10650044

Deng, Y., Bian, J., Wu, S., Lai, J., & Xie, X. (2025). Multiplex graph aggregation and feature refinement for unsupervised incomplete multimodal emotion recognition. *Information Fusion*, *114*, 102711. https://doi.org/10.1016/J.INFFUS.2024.102711

Ding, J.-E., Hsu, C.-C., & Liu, F. (2024). Parkinson's disease classification using contrastive graph cross-view learning with multimodal fusion of spect images and clinical features. *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*, 1–5. https://doi.org/10.1109/ISBI56570.2024.10635712

Ding, K., Zhou, M., Wang, Z., Liu, Q., Arnold, C. W., Zhang, S., & Metaxas, D. N. (2022). Graph convolutional networks for multi-modality medical imaging: Methods, architectures, and clinical applications. https://doi.org/10.48550/arXiv.2202.08916

D'Souza, N. S., Wang, H., Giovannini, A., Foncubierta-Rodriguez, A., Beck, K. L., Boyko, O., & Syeda-Mahmood, T. F. (2024). Fusing modalities by multiplexed graph neural networks for outcome prediction from medical data and beyond. *Medical Image Analysis*, *93*, 103064. https://doi.org/https://doi.org/10.1016/j.media.2023.103064

Duvieusart, B., Krones, F., Parsons, G., Tarassenko, L., Papiez, B. W., & Mahdi, A. (2022). Multimodal cardiomegaly classification with image-derived digital biomarkers. In G.-Z. Yang et al. (Eds.), *Medical image understanding and analysis (miua) 2022* (pp. 13–27, Vol. 13413). Springer Nature Switzerland AG. https://doi.org/10.1007/978-3-031-12053-4_2

Dwivedi, C., Nofallah, S., Pouryahya, M., Iyer, J., Leidal, K., Chung, C., Watkins, T., Billin, A., Myers, R., Abel, J., & Behrooz, A. (2022). Multi stain graph fusion for multimodal integration in pathology. *IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops*, *2022-June*, 1834–1844. https://doi.org/10.1109/CVPRW56347.2022.00200

Fan, F., Su, Y., Liu, Y., Jing, P., Qu, K., & Liu, Y. (2024). Multimodal deep hierarchical semantic-aligned matrix factorization method for micro-video multi-label classification. *Information Processing & Management*, *61*(5), 103798. https://doi.org/10.1016/J.IPM.2024.103798

Fang, X., Liu, Z., & Xu, M. (2020). Ensemble of deep convolutional neural networks based multi-modality images for alzheimer's disease diagnosis. *IET Image Processing*, *14*, 318–326. https://doi.org/10.1049/iet-ipr.2019.0617

Feng, M., Wang, J., Wen, K., & Sun, J. (2023). Grading of diabetic retinopathy images based on graph neural network. *IEEE Access*, *11*, 98391–98401. https://doi.org/10.1109/ACCESS.2023.3312709

Fout, A., Byrd, J., Shariat, B., & Ben-Hur, A. (2017). Protein interface prediction using graph convolutional networks. *31st Conference on Neural Information Processing Systems (NIPS)*. https://doi.org/10.5555/3295222.3295399

Fu, C., Qian, F., Su, K., Su, Y., Wang, Z., Shi, J., Liu, Z., Liu, C., & Ishi, C. T. (2025). HiMul-LGG: A hierarchical decision fusion-based local–global graph neural network for multimodal emotion recognition in conversation. *Neural Networks*, *181*, 106764. https://doi.org/10.1016/J.NEUNET.2024.106764

Gao, J., Zhao, Y., & . . . (2025). Graph neural networks and multimodal dti features for schizophrenia classification: Insights from brain network analysis and gene expression. *Neuroscience Bulletin*, *41*, 1–15. https://doi.org/10.1007/s12264-025-01385-5

Gao, J., Li, P., Chen, Z., & Zhang, J. (2020). A survey on deep learning for multimodal data fusion. *Neural Computation*, *32*(5), 829–864.

Gao, J., Qian, M., Wang, Z., Li, Y., Luo, N., Xie, S., Shi, W., Li, P., Chen, J., Chen, Y., Wang, H., Liu, W., Li, Z., Yang, Y., Guo, H., Wan, P., Lv, L., Lu, L., Yan, J., . . . Jiang, T. (2024). Exploring schizophrenia classification through multimodal mri and deep graph neural networks: Unveiling brain region-specific weight discrepancies and their association with cell-type specific transcriptomic features. *Schizophrenia Bulletin*, *51*(1), 217–235. https://doi.org/10.1093/schbul/sbae069

Gruden, J. F., Steinberger, S., & Green, D. B. (2025). Micronodular lung disease. In *Diseases of the chest, heart and vascular system 2025–2028* (pp. 159–167). Springer. https://doi.org/10.1007/978-3-031-83872-9_13

Guo, S., Wang, L., Chen, Q., Wang, L., Zhang, J., & Zhu, Y. (2022). Multimodal MRI image decision fusion-based network for glioma classification. *Frontiers in Oncology*, *12*, 819673. https://doi.org/10.3389/fonc.2022.819673

Hamilton, W. L., Ying, R., & Leskovec, J. (2017). Inductive representation learning on large graphs. *Advances in Neural Information Processing Systems (NeurIPS)*, *30*.

Han, K., Wang, Y., Guo, J., Tang, Y., & Wu, E. (2022). Vision gnn: An image is worth graph of nodes. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, & A. Oh (Eds.), *Advances in neural information processing systems* (pp. 8291–8303, Vol. 35). Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2022/file/3743e69c8e47eb2e6d3afaea80e439fb-Paper-Conference.pdf

Hao, D., & Li, H. (2023). A graph-based edge attention gate medical image segmentation method. *IET Image Processing*, *17*, 2142–2157. https://doi.org/10.1049/ipr2.12780

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778. https://doi.org/10.1109/CVPR.2016.90

He, M., Han, K., Zhang, Y., & Chen, W. (2021). Hierarchical-order multimodal interaction fusion network for grading gliomas. *Physics in Medicine & Biology*, *66*(21), 215016. https://doi.org/10.1088/1361-6560/AC30A1

He, R., Jiao, G., & Li, C. (2024). ETBHD-HMF: A Hierarchical Multimodal Fusion Architecture for Enhanced Text-Based Hair Design. *Computer Graphics Forum*, *43*(6), e15194. https://doi.org/10.1111/CGF.15194

Huang, B., Yang, F., Yin, M., Mo, X., & Zhong, C. (2020). A review of multimodal medical image fusion techniques. *Computational and Mathematical Methods in Medicine*, *2020*, 8279342. https://doi.org/10.1155/2020/8279342

Huang, S.-C., Pareek, A., Seyyedi, S., Banerjee, I., & Lungren, M. P. (2020). Fusion of medical imaging and electronic health records using deep learning: A systematic review and implementation guidelines. *npj Digital Medicine*, *3*(1), 1–9.

Hyup, H. S., Sun, A. M., Wha, K. Y., & Jin, L. S. (2019). A machine-learning approach using pet-based radiomics to predict the histological subtypes of lung cancer. *Clinical Nuclear Medicine*, *44*(12), 956–960. https://doi.org/10.1097/RLU.0000000000002810

Jaume, G., Pati, P., Foncubierta-Rodriguez, A., Feroce, F., Scognamiglio, G., Anniciello, A. M., Thiran, J.-P., Goksel, O., & Gabrani, M. (2020). Towards explainable graph representations in digital pathology. https://arxiv.org/abs/2007.00311v1

Jia, H., Tang, H., Ma, G., Cai, W., Huang, H., Zhan, L., & Xia, Y. (2023). A convolutional neural network with pixel-wise sparse graph reasoning for covid-19 lesion segmentation in ct images. *Computers in Biology and Medicine*, *155*, 106698. https://doi.org/10.1016/j.compbiomed.2023.106698

Jiang, L., & Meng, Z. (2023). Knowledge-Based Visual Question Answering Using Multi-Modal Semantic Graph. *Electronics 2023, Vol. 12, Page 1390*, *12*(6), 1390. https://doi.org/10.3390/ELECTRONICS12061390

Jie, C., Jiming, C., Ying, S., Yanchun, T., & Haodong, R. (2024). A pyramid gnn model for cxr-based covid-19 classification. *The Journal of Supercomputing*, *80*, 5490–5508. https://doi.org/10.1007/s11227-023-05633-1

Jin, K., Wang, X., Wang, L., Guo, W., Han, Q., & Yu, X. (2025). Multimodal medical image fusion based on dilated convolution and attention-based graph convolutional network. *Computers and Electrical Engineering*, *124*, 110359. https://doi.org/https://doi.org/10.1016/j.compeleceng.2025.110359

Kazi, A., Krishna, S. A., Shekarforoush, S., Kortuem, K., Albarqouni, S., & Navab, N. (2019). Self-attention equipped graph convolutions for disease prediction. *International Symposium on Biomedical Imaging*, 1896–1899. https://doi.org/10.1109/ISBI.2019.8759274

Keicher, M., Burwinkel, H., Bani-Harouni, D., et al. (2023). Multimodal graph attention network for covid-19 outcome prediction. *Scientific Reports*, *13*, 19539. https://doi.org/10.1038/s41598-023-46625-8

Kermany, D., Zhang, K., & Goldbaum, M. (2018). *Labeled optical coherence tomography (oct) and chest x-ray images for classification* (Version 2). Mendeley Data. https://doi.org/10.17632/rscbjbr9sj.2

Khattar, A., & Quadri, S. M. (2022). CAMM: Cross-Attention Multimodal Classification of Disaster-Related Tweets. *IEEE Access*, *10*, 92889–92902. https://doi.org/10.1109/ACCESS.2022.3202976

Khatun, Z., Jónsson, H., Tsirilaki, M., Maffulli, N., Oliva, F., Daval, P., Tortorella, F., & Gargiulo, P. (2024). Beyond pixel: Superpixel-based mri segmentation through traditional machine learning and graph convolutional network. *Computer Methods and Programs in Biomedicine*, *256*, 108398. https://doi.org/https://doi.org/10.1016/j.cmpb.2024.108398

Khemani, B., Patil, S., Kotecha, K., & Tanwar, S. (2024). A review of graph neural networks: Concepts, architectures, techniques, challenges, datasets, applications, and future directions. *Journal of Big Data*, *11*(1), 1–43. https://doi.org/10.1186/S40537-023-00876-4

Kim, S., Lee, N., Lee, J., Hyun, D., & Park, C. (2023). Heterogeneous graph learning for multi-modal medical data analysis. *Proceedings of the AAAI Conference on Artificial Intelligence*, 5141–5150. https://doi.org/10.1609/aaai.v37i4.25643

Kipf, T. N., & Welling, M. (2017). Semi-supervised classification with graph convolutional networks. https://arxiv.org/abs/1609.02907

Kong, Z., Zhou, R., Luo, X., Zhao, S., Ragin, A. B., Leow, A. D., & He, L. (2024). TGNet: tensor-based graph convolutional networks for multimodal brain network analysis. *BioData Mining*, *17*(55). https://doi.org/10.1186/s13040-024-00409-6

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. *Advances in Neural Information Processing Systems*, *25*, 1097–1105. http://code.google.com/p/cuda-convnet/

Lan, W., Hao, J., Zhou, S., Zhang, J., Ma, S., & Zhao, Y. (2024). Hybrid graph representation learning for carotid artery stenosis detection based on multimodal retinal OCTA images. *IEEE Access*. https://doi.org/10.1109/ACCESS.2024.3412961

Lee, G.-B., Jeong, Y.-J., Kang, D.-Y., Yun, H.-J., & Yoon, M. (2024). Multimodal feature fusion-based graph convolutional networks for alzheimer's disease stage classification using f-18 florbetaben brain pet images and clinical indicators. *PLOS ONE*, *19*(12), e0315809. https://doi.org/10.1371/journal.pone.0315809

Li, H., & Fan, Y. (2019). Early prediction of alzheimer's disease dementia based on baseline hippocampal mri and 1-year follow-up cognitive measures using deep recurrent neural networks. *Proceedings of the 2019 IEEE International Symposium on Biomedical Imaging (ISBI)*, 368–371.

Li, J., Wang, X., Lv, G., & Zeng, Z. (2023). GraphMFT: A graph network based multimodal fusion technique for emotion recognition in conversation. *Neurocomputing*, *550*, 126427. https://doi.org/10.1016/J.NEUCOM.2023.126427

Li, K., Han, S., Yang, L., Sun, Z., Yu, Z., Xu, H., Ma, L., Gao, J., & Jiang, H. (2023). Msa-gcn: A multi-information selection aggregation graph convolutional network for breast tumor grading. *IEEE Journal of Biomedical and Health Informatics*, *27*, 5994–6005. https://doi.org/10.1109/JBHI.2023.3327755

Li, R., Wu, X., Li, A., & Wang, M. (2022). HFBSurv: hierarchical multimodal fusion with factorized bilinear models for cancer survival prediction. *Bioinformatics*, *38*(9), 2587–2594. https://doi.org/10.1093/BIOINFORMATICS/BTAC113

Li, T., Zhang, Y., Su, D., Liu, M., Ge, M., Chen, L., Li, C., & Tang, J. (2025). Knowledge graph-based few-shot learning for label of medical imaging reports. *Academic Radiology*. https://doi.org/https://doi.org/10.1016/j.acra.2025.02.045

Li, W., Zhao, J., Shen, C., Zhang, J., Hu, J., Xiao, M., Zhang, J., & Chen, M. (2022). Regional brain fusion: Graph convolutional network for alzheimer's disease prediction and analysis. *Frontiers in Neuroinformatics*, *Volume 16 - 2022*. https://doi.org/10.3389/fninf.2022.886365

Li, X., Zhou, Y., Dvornek, N., Zhang, M., Gao, S., Zhuang, J., Scheinost, D., Staib, L. H., Ventola, P., & Duncan, J. S. (2021). Braingnn: Interpretable brain graph neural network for fmri analysis. *Medical Image Analysis*, *74*, 102233. https://doi.org/https://doi.org/10.1016/j.media.2021.102233

Li, X., Zhang, L., Zhao, Q., Zhu, Z., & Zhang, F. (2025). P2gcn: Pixel-patch mutual enhancement graph convolutional network for sonar image super-resolution. *Expert Systems with Applications*, *279*, 127265. https://doi.org/https://doi.org/10.1016/j.eswa.2025.127265

Li, Y., El Habib Daho, M., Conze, P. H., Al Hajj, H., Bonnin, S., Ren, H., Manivannan, N., Magazzeni, S., Tadayoni, R., Cochener, B., Lamard, M., & Quellec, G. (2022). Multimodal Information Fusion for Glaucoma and Diabetic Retinopathy Classification. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, *13576 LNCS*, 53–62. https://doi.org/10.1007/978-3-031-16525-2{_}6

Li, Y., El Habib Daho, M., Conze, P. H., Zeghlache, R., Le Boité, H., Tadayoni, R., Cochener, B., Lamard, M., & Quellec, G. (2024). A review of deep learning-based information fusion techniques for multimodal medical image classification. *Computers in Biology and Medicine*, *177*, 108635. https://doi.org/10.1016/J.COMPBIOMED.2024.108635

Li, Y., Liu, S., Wang, X., & Jing, P. (2023). Self-supervised deep partial adversarial network for micro-video multimodal classification. *Information Sciences*, *630*, 356–369. https://doi.org/10.1016/J.INS.2022.11.111

Lian, R., & Ling, H. (2023). Checkerpose: Progressive dense keypoint localization for object pose estimation with graph neural network [arXiv preprint arXiv:2303.16874]. https://arxiv.org/abs/2303.16874

Lin, L., Xiong, M., Zhang, G., Kang, W., Sun, S., Wu, S., & Neuroimaging, I. A. D. (2023). A convolutional neural network and graph convolutional network based framework for ad classification. *Sensors*, *23*(4), 1914. https://doi.org/10.3390/s23041914

Liu, F., Yuan, S., Li, W., Xu, Q., & Sheng, B. (2023). Patch-based deep multimodal learning framework for alzheimer's disease diagnosis using multi-view neuroimaging. *Biomedical Signal Processing and Control*, *80*, 104400. https://doi.org/https://doi.org/10.1016/j.bspc.2022.104400

Liu, K., Wang, J., & Zhang, X. (2024). Debiased momentum contrastive learning for multimodal video similarity measures. *Neurocomputing*, *563*, 126938. https://doi.org/10.1016/J.NEUCOM.2023.126938

Liu, X., Zhen, J., Liu, Y., Geng, G., & Shui, W. (2025). Multiple channel gcn with multiple directional gaussian for porcelain microscopic image classification. *npj Heritage Science*, *13*, 254. https://doi.org/10.1038/s40494-025-01716-9

Lu, P., Hu, L., Mitelpunkt, A., Bhatnagar, S., Lu, L., & Liang, H. (2024). A hierarchical attention-based multimodal fusion framework for predicting the progression of Alzheimer's disease. *Biomedical Signal Processing and Control*, *88*, 105669. https://doi.org/10.1016/J.BSPC.2023.105669

McConnell, N., et al. (2026). A computationally frugal, open-source chest ct foundation model for thoracic disease detection in lung cancer screening programmes. *Communications Medicine*, *6*, 24. https://doi.org/10.1038/s43856-025-01328-1

Nash, C., Nair, R., & Naqvi, S. M. (2024). Cross-Modal Attention for Multimodal Information Fusion: A Novel Approach to Attention Deficit Hyperactivity Disorder Detection. *FUSION 2024 - 27th International Conference on Information Fusion*. https://doi.org/10.23919/FUSION59988.2024.10706381

Nie, D., Lu, J., Zhang, H., Adeli, E., Wang, J., Yu, Z., Liu, L., Wang, Q., Wu, J., & Shen, D. (2019). Multi-channel 3d deep feature learning for survival time prediction of brain tumor patients using multi-modal neuroimages. *Scientific Reports*, *9*(1), 1–14.

Palmal, S., Arya, N., Saha, S., & Tripathy, S. (2024). Integrative prognostic modeling for breast cancer: Unveiling optimal multimodal combinations using graph convolutional networks and calibrated random forest. *Applied Soft Computing*, *154*, 111379. https://doi.org/10.1016/J.ASOC.2024.111379

Pan, J., Lin, H., Dong, Y., Wang, Y., & Ji, Y. (2022). MAMF-GCN: Multi-scale adaptive multi-channel fusion deep graph convolutional network for predicting mental disorder. *Computers in Biology and Medicine*, *148*, 105823. https://doi.org/10.1016/j.compbiomed.2022.105823

Parisot, S., Ktena, S. I., Ferrante, E., Lee, M., Guerrero, R., Glocker, B., & Rueckert, D. (2018). Disease prediction using graph convolutional networks: Application to autism spectrum disorder and alzheimer's disease. *Medical Image Analysis*, *48*, 117–130. https://doi.org/10.1016/J.MEDIA.2018.06.001

Patel, P. (2020). Chest x-ray (covid-19 & pneumonia) [Available at: https://www.kaggle.com/datasets/prashant268/chest-xray-covid19-pneumonia].

Paul, S., Yener, B., & Lund, A. W. (2024). C2p-gcn: Cell-to-patch graph convolutional network for colorectal cancer grading. *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 1–4. https://doi.org/10.1109/EMBC53108.2024.10782435

Peng, C., He, J., & Xia, F. (2024). Learning on multimodal graphs: A survey. *arXiv preprint arXiv:2402.05322*. https://arxiv.org/abs/2402.05322

Peng, L., Cai, S., Wu, Z., Shang, H., Zhu, X., & Li, X. (2024). Mmgpl: Multimodal medical data analysis with graph prompt learning. *Medical Image Analysis*, *97*, 103225. https://doi.org/10.1016/j.media.2024.103225

Pingi, S. T., Zhang, D., Bashar, M. A., & Nayak, R. (2024). Joint Representation Learning with Generative Adversarial Imputation Network for Improved Classification of Longitudinal Data. *Data Science and Engineering*, *9*(1), 5–25. https://doi.org/10.1007/S41019-023-00232-9/TABLES/7

Pölsterl, S., Wolf, T. N., & Wachinger, C. (2021). Combining 3d image and tabular data via the dynamic affine feature map transform. In M. de Bruijne, P. C. Cattin, S. Cotin, N. Padoy, S. Speidel, Y. Zheng, & C. Essert (Eds.), *Medical image computing and computer assisted intervention – miccai 2021* (pp. 688–698, Vol. 12905). Springer, Cham. https://doi.org/10.1007/978-3-030-87240-3_66

Pradhan, R., & Santosh, K. C. (2024). Analyzing pulmonary abnormality with superpixel based graph neural network in chest X-Ray. In K. C. Santosh, A. Makkar, M. Conway, A. K. Singh, A. Vacavant, A. A. el Kalam, M.-R. Bouguelia, & R. Hegadi (Eds.), *Recent trends in image processing and pattern recognition* (pp. 97–110). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-53085-2_9

Qi, X., Wen, Y., Zhang, P., & Huang, H. (2025). MFGCN: Multimodal fusion graph convolutional network for speech emotion recognition. *Neurocomputing*, *611*, 128646. https://doi.org/10.1016/J.NEUCOM.2024.128646

Qian, X., Zhang, B., Liu, S., Wang, Y., Chen, X., Liu, J., Yang, Y., Chen, X., Wei, Y., Xiao, Q., Ma, J., Shung, K. K., Zhou, Q., Liu, L., & Chen, Z. (2020). A combined ultrasonic b-mode and color doppler system for the classification of breast masses using neural network. *European Radiology*, *30*(5), 3023–3033. https://doi.org/10.1007/S00330-019-06610-0

Qin, R., Wang, Z., Jiang, L., Qiao, K., Hai, J., Chen, J., Xu, J., Shi, D., & Yan, B. (2020). Fine-grained lung cancer classification from pet and ct images based on multidimensional attention mechanism. *Complexity*, *2020*(1), 6153657. https://doi.org/10.1155/2020/6153657

Qiu, L., Zhao, L., Hou, R., Zhao, W., Zhang, S., Lin, Z., Teng, H., & Zhao, J. (2023). Hierarchical multimodal fusion framework based on noisy label learning and attention mechanism for cancer classification with pathology and genomic features. *Computerized Medical Imaging and Graphics*, *104*, 102176. https://doi.org/10.1016/J.COMPMEDIMAG.2022.102176

Qiu, S., Chang, G. H., Panagia, M., Gopal, D. M., Au, R., & Kolachalama, V. B. (2018). Fusion of deep learning models of mri scans, mini–mental state examination, and logical memory test enhances diagnosis of mild cognitive impairment. In *Alzheimer's & dementia: Diagnosis, assessment & disease monitoring* (pp. 737–749, Vol. 10). Elsevier.

Rahaman, M. M., Millar, E. K., & Meijering, E. (2024). Generalized deep learning for histopathology image classification using supervised contrastive learning. *Journal of Advanced Research*. https://doi.org/https://doi.org/10.1016/j.jare.2024.11.013

Rahimi, N., Kumar, C., McLinden, J., Ismail Hosni, S., Bahram Borgheai, S., Shahriari, Y., & Shao, M. (2024). Topology-Aware Multimodal Fusion for Neural Dynamics Representation Learning and Classification. *IEEE Sensors Journal*, *24*(13), 21062–21073. https://doi.org/10.1109/JSEN.2024.3400006

Rahman, T., et al. (2020). Covid-19 radiography database [Available at: https://www.kaggle.com/datasets/tawsifurrahman/covid19-radiography-database].

Rajeshwari, R., & Geetha, R. (2025). Twin-layer attention graph convolutional network (tla-gcn): Enhancing abnormality detection in chest x-rays. *Applied Soft Computing*, *181*, 113514. https://doi.org/10.1016/j.asoc.2025.113514

Rayed, M. E., Islam, S. S., Niha, S. I., Jim, J. R., Kabir, M. M., & Mridha, M. (2024). Deep learning for medical image segmentation: State-of-the-art advancements and challenges. *Informatics in Medicine Unlocked*, *47*, 101504. https://doi.org/10.1016/j.imu.2024.101504

Reda, I., Khalil, A., Elmogy, M., Abou El-Fetouh, A., Shalaby, A., Abou El-Ghar, M., Elmaghraby, A., Ghazal, M., & El-Baz, A. (2018). Deep learning role in early diagnosis of prostate cancer. *Technology in Cancer Research & Treatment*, *17*, 1533034618775530. https://doi.org/10.1177/1533034618775530

Renton, G., Balcilar, M., Héroux, P., Gaüzère, B., Honeine, P., & Adam, S. (2021). Symbols detection and classification using graph neural networks [Special Issue on Graph-Based Pattern Recognition]. *Pattern Recognition Letters*, *152*, 391–397. https://doi.org/10.1016/j.patrec.2021.09.020

Salekin, M. S., Zamzmi, G., Goldgof, D., Kasturi, R., Ho, T., & Sun, Y. (2021). Multimodal spatiotemporal deep learning approach for neonatal postoperative pain assessment. *Comput. Biol. Med.*, *129*, 104150. https://doi.org/10.1016/j.compbiomed.2020.104150

Sankar, A., Wu, Y., Gou, L., Zhang, W., & Yang, H. (2018). Dynamic graph representation learning via self-attention networks. https://arxiv.org/abs/1812.09430

Screven, T., Necz, A., Smucny, J., & Davidson, I. (2025). Using graph convolutional networks to address fmri small data problems. https://arxiv.org/abs/2502.17489

Searle, T., Ibrahim, Z., & Dobson, R. (2020). Comparing natural language processing techniques for alzheimer's dementia prediction in spontaneous speech. *Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH*, *2020-October*, 2192–2196. https://doi.org/10.21437/Interspeech.2020-2729

Shaban, A., & Yousefi, S. (2024). Multimodal deep learning. *Springer Optimization and Its Applications*, *211*, 209–219. https://doi.org/10.1007/978-3-031-53092-0_10/TABLES/1

Shamshad, F., Khan, S., Zamir, S. W., Khan, M. H., Hayat, M., Khan, F. S., & Fu, H. (2023). Transformers in medical imaging: A survey. *Medical Image Analysis*, *88*, 102802. https://doi.org/10.1016/j.media.2023.102802

Shen, Y., Guo, J., Liu, Y., Xu, C., Li, Q., & Qi, F. (2025). Smanet: Superpixel-guided multi-scale attention network for medical image segmentation. *Biomedical Signal Processing and Control*, *100*, 107062. https://doi.org/https://doi.org/10.1016/j.bspc.2024.107062

Shi, G., Zhu, Y., Liu, W., Yao, Q., & Li, X. (2022). Heterogeneous graph-based multimodal brain network learning. https://arxiv.org/abs/2110.08465

Shin, S. Y., Lee, S., Yun, I. D., & Lee, K. M. (2019). Deep vessel segmentation by learning graphical connectivity. *Medical Image Analysis*, *58*, 101556. https://doi.org/10.1016/j.media.2019.101556

Shovon, I. I., Ahmad, I., & Shin, S. (2025). Segmentation aided multiclass classification of lung disease in chest x-ray images using graph neural networks. *2025 International Conference on Information Networking (ICOIN)*, 666–669. https://doi.org/10.1109/ICOIN63865.2025.10992753

Shuyin, X., Dawei, D., Long, Y., Li, Z., Danf, L., hao, Z., & Guoy, W. (2023). Graph-based representation for image based on granular-ball. https://arxiv.org/abs/2303.02388

Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. *3rd International Conference on Learning Representations, ICLR 2015 - Conference Track Proceedings*. https://arxiv.org/abs/1409.1556v6

Soares, E., Angelov, P., Biaso, S., Froes, M. H., & Abe, D. K. (2020). Sars-cov-2 ct-scan dataset: A large dataset of real patients ct scans for sars-cov-2 identification. *medRxiv*. https://doi.org/10.1101/2020.04.24.20078584

Song, X., Zhou, F., Frangi, A. F., Cao, J., Xiao, X., Lei, Y., Wang, T., & Lei, B. (2021). Graph convolution network with similarity awareness and adaptive calibration for disease-induced deterioration prediction. *Medical Image Analysis*, *69*, 101947. https://doi.org/10.1016/J.MEDIA.2020.101947

Sun, J., Xue, J., Xu, Z., Li, N., Tang, C., Zhao, L., Pu, B., & Zhang, Y. (2025). Sagcn: Self-adaptive graph convolutional network for pneumonia detection. *Biomedical Signal Processing and Control*, *106*, 107634. https://doi.org/https://doi.org/10.1016/j.bspc.2025.107634

Syed, Z. S., Syed, M. S. S., Lech, M., & Pirogova, E. (2021). Automated recognition of alzheimer's dementia using bag-of-deep-features and model ensembling. *IEEE Access*, *9*, 88377–88390. https://doi.org/10.1109/ACCESS.2021.3090321

Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., & Rabinovich, A. (2015). Going deeper with convolutions. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, *07-12-June-2015*, 1–9. https://doi.org/10.1109/CVPR.2015.7298594

Tang, C., Hu, C., Sun, J., Wang, S.-H., & Zhang, Y.-D. (2022). Nscgcn: A novel deep gcn model to diagnosis covid-19. *Computers in Biology and Medicine*, *150*, 106151. https://doi.org/https://doi.org/10.1016/j.compbiomed.2022.106151

Tang, C., Wei, M., Sun, J., Wang, S., & Zhang, Y. (2023). Csagp: Detecting alzheimer's disease from multimodal images via dual-transformer with cross-attention and graph pooling. *Journal of King Saud University - Computer and Information Sciences*, *35*(7), 101618. https://doi.org/https://doi.org/10.1016/j.jksuci.2023.101618

Tariq, M., Weichwald, S., Janke, A., Mueller, S., Scherer, S., Fuest, M., Schopf, F., Liew, C., Zhang, Q., Martin, S., et al. (2023). Graph-based fusion modeling and explanation for disease trajectory prediction. *NPJ Digital Medicine*, *6*(1), 27.

Tian, Y., Chen, H., Xu, C., & Wang, Y. (2024). Image processing GNN: Breaking rigidity in super-resolution. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 24108–24117. https://doi.org/10.1109/CVPR52733.2024.02276

Todescato, M. V., Garcia, L. F., Balreira, D. G., & Carbonera, J. L. (2024). Multiscale patch-based feature graphs for image classification. *Expert Systems with Applications*, *235*, 121116. https://doi.org/10.1016/j.eswa.2023.121116

Tsai, Y. H. H., Liang, P. P., Zadeh, A., Morency, L. P., & Salakhutdinov, R. (2018). Learning factorized multimodal representations. *7th International Conference on Learning Representations, ICLR 2019*. https://arxiv.org/abs/1806.06176v3

Vale-Silva, L. A., & Rohr, K. (2020). Pan-cancer prognosis prediction using multimodal deep learning. *Proceedings of the 2020 IEEE International Symposium on Biomedical Imaging (ISBI)*, 568–571.

Vanguri, R. S., Luo, J., Aukerman, A. T., Egger, J. V., Fong, C. J., Horvat, N., Pagano, A., de Arimateia Batista Araujo-Filho, J., Geneslaw, L., Rizvi, H., et al. (2022). Multimodal integration of radiology, pathology and genomics for prediction of response to pd-(l)1 blockade in patients with non-small cell lung cancer. *Nature Cancer*, *3*(10), 1151–1164.

Vasudevan, V., Bassenne, M., Islam, M. T., & Xing, L. (2023). Image classification using graph neural network and multiscale wavelet superpixels. *Pattern Recognition Letters*, *166*, 89–96. https://doi.org/10.1016/j.patrec.2023.01.003

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems*, *2017-December*, 5999–6009. https://arxiv.org/abs/1706.03762v7

Veličković, P., Casanova, A., Liò, P., Cucurull, G., Romero, A., & Bengio, Y. (2017). Graph attention networks. *6th International Conference on Learning Representations, ICLR 2018 - Conference Track Proceedings*. https://doi.org/10.1007/978-3-031-01587-8_7

Walker, B., Krones, F., Kiskin, I., Parsons, G., Lyons, T., & Mahdi, A. (2022). Dual bayesian resnet: A deep learning approach to heart murmur detection. *Computers in Cardiology*.

Wang, B., Pan, H., Aboah, A., Zhang, Z., Keles, E., Torigian, D., Turkbey, B., Krupinski, E., Udupa, J., & Bagci, U. (2024). Gazegnn: A gaze-guided graph neural network for chest x-ray classification. *2024 IEEE/CVF Winter Conference on*

Applications of Computer Vision (WACV), 2183–2192. https://doi.org/10.1109/WACV57701.2024.00219

Wang, R., Guo, W., Wang, Y., Zhou, X., Leung, J. C., Yan, S., & Cui, L. (2024). Hybrid multimodal fusion for graph learning in disease prediction. *Methods*, *229*, 41–48. https://doi.org/10.1016/J.YMETH.2024.06.003

Wang, T., Shao, W., Huang, Z., Tang, H., Zhang, J., Ding, Z., & Huang, K. (2021). MOGONET integrates multi-omics data using graph convolutional networks allowing patient classification and biomarker identification. *Nature Communications 2021 12:1*, *12*(1), 1–13. https://doi.org/10.1038/s41467-021-23774-w

Wang, Z., Ma, J., Gao, Q., Bain, C., Imoto, S., Liò, P., Cai, H., Chen, H., & Song, J. (2024). Dual-stream multi-dependency graph neural network enables precise cancer survival analysis. *Medical Image Analysis*, *97*, 103252. https://doi.org/10.1016/j.media.2024.103252

Waqas, A., Tripathi, A., Ramachandran, R. P., Stewart, P., & Rasool, G. (2023). Multimodal data integration for oncology in the era of deep neural networks: A review. *Frontiers in Artificial Intelligence*, *7*. https://doi.org/10.3389/frai.2024.1408843

Weers, A., Berger, A. H., Lux, L., Schüffler, P., Rueckert, D., & Paetzold, J. C. (2025). From pixels to histopathology: A graph-based framework for interpretable whole slide image analysis. https://arxiv.org/abs/2503.11846

Wu, J., Ma, J., Xi, H., Li, J., & Zhu, J. (2025). Multi-scale graph harmonies: Unleashing u-net's potential for medical image segmentation through contrastive learning. *Neural Networks*, *182*, 106914. https://doi.org/https://doi.org/10.1016/j.neunet.2024.106914

Xia, Y., Liu, L., Dong, T., Chen, J., Cheng, Y., & Tang, L. (2024). A depression detection model based on multimodal graph neural network. *Multimedia Tools and Applications*, *83*, 63379–63395. https://doi.org/10.1007/s11042-023-18079-7

Xie, C., Zhong, W., Xu, W., & Mueller, K. (2019). Visual analytics of heterogeneous data using hypergraph learning. *ACM Transactions on Intelligent Systems and Technology*, *10*(1). https://doi.org/10.1145/3200765

Xu, H., Tang, W., Li, Z., Qin, K., & Zou, J. (2024). Multimodal Dual Cross-Attention Fusion Strategy for Autonomous Garbage Classification System. *IEEE Transactions on Industrial Informatics*. https://doi.org/10.1109/TII.2024.3435508

Xu, K., Li, C., Tian, Y., Sonobe, T., Kawarabayashi, K., & Jegelka, S. (2018). Representation learning on graphs with jumping knowledge networks. *Proc. Int. Conf. Mach. Learn*, 5453–5462. https://proceedings.mlr.press/v80/xu18c.html

Xu, X., Jiang, X., Ma, C., Du, P., Li, X., Lv, S., Yu, L., Ni, Q., Chen, Y., Su, J., Lang, G., Li, Y., Zhao, H., Liu, J., Xu, K., Ruan, L., Sheng, J., Qiu, Y., Wu, W., ... Li, L. (2020). A deep learning system to screen novel coronavirus disease 2019 pneumonia. *Engineering*, *6*(10), 1122–1129. https://doi.org/https://doi.org/10.1016/j.eng.2020.04.010

Xu, Y., Quan, R., Xu, W., Huang, Y., Chen, X., & Liu, F. (2024). Advances in medical image segmentation: A comprehensive review of traditional, deep learning and hybrid approaches. *Bioengineering*, *11*(10). https://doi.org/10.3390/bioengineering11101034

Yang, Y., Guo, X., Chang, Z., Ye, C., Xiang, Y., & Ma, T. (2022). Multi-modal dynamic graph network: Coupling structural and functional connectome for disease diagnosis and classification. https://arxiv.org/abs/2210.13721

Ye, J., Zeng, A., Pan, D., Chen, J., & Cheng, G. (2025). Medgnn: General medical image recognition network via gnn visual representations. In J. C. Gee, D. C. Alexander, J. Hong, J. E. Iglesias, C. H. Sudre, A. Venkataraman, P. Golland, J. H. Kim, & J. Park (Eds.), *Medical image computing and computer assisted intervention – miccai 2025* (pp. 332–342). Springer Nature Switzerland.

Yen, C., Lin, C. L., & Chiang, M. C. (2023). Exploring the frontiers of neuroimaging: A review of recent advances in understanding brain functioning and disorders. *Life*, *13*(7), 1472. https://doi.org/10.3390/LIFE13071472

Yue, Z., Ding, S., Yang, S., Wang, L., & Li, Y. (2022). Multimodal Information Fusion Approach for Noncontact Heart Rate Estimation Using Facial Videos and Graph Convolutional Network. *IEEE Transactions on Instrumentation and Measurement*, *71*, 1–13. https://doi.org/10.1109/TIM.2021.3129498

Zaidi, S. A. M., Latif, S., & Qadir, J. (2024). Enhancing Cross-Language Multimodal Emotion Recognition With Dual Attention Transformers. *IEEE Open Journal of the Computer Society*, *5*, 684–3. https://doi.org/10.1109/OJCS.2024.3486904

Zhang, D., Nayak, R., & Bashar, M. A. (2024). Pre-gating and contextual attention gate — A new fusion method for multi-modal data tasks. *Neural Networks*, *179*, 106553. https://doi.org/10.1016/J.NEUNET.2024.106553

Zhang, L., Zhang, J., Li, Z., & Xu, J. (2020). Towards Better Graph Representation: Two-Branch Collaborative Graph Neural Networks For Multimodal Marketing Intention Detection. *2020 IEEE International Conference on Multimedia and Expo (ICME)*, 1–6. https://doi.org/10.1109/ICME46284.2020.9102918

Zhang, M., & Chen, Y. (2018). Link prediction based on graph neural networks. *NIPS'18: Proceedings of the 32nd International Conference on Neural Information Processing Systems*. https://doi.org/10.5555/3327345.3327423

Zhang, M., Cui, Q., Lü, Y., & Li, W. (2024). A feature-aware multimodal framework with auto-fusion for alzheimer's disease diagnosis. *Computers in Biology and Medicine*, *178*, 108740. https://doi.org/10.1016/j.compbiomed.2024.108740

Zhang, T., & Shi, M. (2020). Multi-modal neuroimaging feature fusion for diagnosis of Alzheimer's disease. *Journal of Neuroscience Methods*, *341*, 108795. https://doi.org/10.1016/J.JNEUMETH.2020.108795

Zhang, W., Ding, Q., Hu, J., Ma, Y., & Lu, M. (2023). Pixel-wise graph attention networks for person re-identification. *arXiv preprint arXiv:2307.09183*. https://arxiv.org/abs/2307.09183

Zhang, Y., He, X., Chan, Y. H., Teng, Q., & Rajapakse, J. C. (2023). Multi-modal graph neural network for early diagnosis of alzheimer's disease from smri and pet scans. *Computers in Biology and Medicine*, *164*, 107328. https://doi.org/10.1016/j.compbiomed.2023.107328

Zhang, Y. D., Satapathy, S. C., Guttery, D. S., Górriz, J. M., & Wang, S. H. (2021). Improved breast cancer classification through combining graph convolutional network and convolutional neural network. *Information Processing Management*, *58*(2), 102439. https://doi.org/10.1016/J.IPM.2020.102439

Zhang, Y., Zhao, Y., Wang, M., Dong, Y., Yang, B., Gong, Y., & Feng, X. (2025). A medical image segmentation method based on adaptive graph sparse algorithm under contrastive learning framework. *Displays*, *87*, 102971. https://doi.org/https://doi.org/10.1016/j.displa.2025.102971

Zhao, X., Ma, J., Wang, L., Zhang, Z., Ding, Y., & Xiao, X. (2025). A review of hyperspectral image classification based on graph neural networks. *Artificial Intelligence Review*, *58*, 172. https://doi.org/10.1007/s10462-025-11169-y

Zhao, X., & Du, X. (2025). Breast cancer histopathological image classification based on graph assisted global reasoning. *Journal of Imaging Informatics in Medicine*. https://doi.org/10.1007/s10278-025-01403-y

Zhao, Y., Li, L., Yu, X., Han, K., Duan, J., Liang, D., Chai, N., & Li, Z.-C. (2025). Survgraph: A hybrid-graph attention network for survival prediction using whole slide pathological images in gastric cancer. *Neural Networks*, *189*, 107607. https://doi.org/https://doi.org/10.1016/j.neunet.2025.107607

Zheng, S., Zhu, Z., Liu, Z., Guo, Z., Liu, Y., Yang, Y., & Zhao, Y. (2022). Multi-modal graph learning for disease prediction. *IEEE Transactions on Medical Imaging*, *41*(9), 2207–2216. https://doi.org/10.1109/TMI.2022.3159264

Zhou, F., Wang, T., Zhong, T., & Trajcevski, G. (2022). Identifying user geolocation with hierarchical graph neural networks and explainable fusion. *Information Fusion*, *81*, 1–13. https://doi.org/10.1016/J.INFFUS.2021.11.004

Zhou, J., Zhang, X., Zhu, Z., Lan, X., Fu, L., Wang, H., & Wen, H. (2021). Cohesive multi-modality feature learning and fusion for covid-19 patient severity prediction. *IEEE Transactions on Circuits and Systems for Video Technology*, *32*(5), 2535–2549.

Zhou, P., Jiang, S., Yu, L., Feng, Y., Chen, C., Li, F., Liu, Y., & Huang, Z. (2021). Use of a Sparse-Response Deep Belief Network and Extreme Learning Machine to Discriminate Alzheimer's Disease, Mild Cognitive Impairment, and Normal Controls Based on Amyloid PET/MRI Images. *Frontiers in Medicine*, *7*, 621204. https://doi.org/10.3389/FMED.2020.621204/BIBTEX

Zidan, U., Gaber, M., & Abdelsamea, M. M. (2025). Ipac: Incorporating intra-image patch context into graph neural networks for medical image classification. https://arxiv.org/abs/2510.23504

Zitnik, M., Agrawal, M., & Leskovec, J. (2018). Modeling polypharmacy side effects with graph convolutional networks. *Bioinformatics*, *34*(13), i457–i466. https://doi.org/10.1093/BIOINFORMATICS/BTY294